

AP CSP | Lesson 30: Responsible Computing

Teacher answer key and marking guide

30 teacher marks. Original AP-style practice; this worksheet does not predict an AP score.

Award each written-response point independently. Accept equivalent correct wording and valid alternative methods. For Q5, award its one mark only for both correct choices and no others.

Questions 1-6 | 1 mark each

Q1. Answer: B

B follows the stated reading-only permission. A confuses access with reuse rights. C confuses possession of a copy with ownership. D treats a missing watermark as permission, which the card does not give.

Q2. Answer: D

D includes permission for modification and redistribution. A confuses price with rights. B ignores authorship and license conditions. C contradicts modification being an open-source permission.

Q3. Answer: A

A satisfies both supplied conditions. B omits credit. C fails to indicate the actual modification. D confuses permission to adapt with owning every contribution. This is a simplified classroom card, not complete CC license text.

Q4. Answer: C

C follows the course distinction. A and B erase authorship or ownership. D treats price-free reading as unlimited permission, which is not established by free access alone.

Q5. Answer: B and D

B distinguishes permission compliance from ethical impact. D supports attribution and following stated conditions. A confuses attribution with permission. C ignores differences among licenses.

Q6. Answer: B

B preserves the distinction between modeled eligibility and actual behavior. A assumes participation is guaranteed. C assumes access repairs every selection or response bias. D ignores unresolved skills and accessibility barriers.

Written-response answers | Questions 7-12

Q7. 4 marks

Answer: Permission determines whether the planned copying and publication are allowed. Attribution identifies the source and creator. Credit alone cannot expand reading-only permission. The team should obtain copying permission before publishing or replace the figure with material whose terms permit the use.

1 mark: Explains that permission must cover the intended copying or publication.

1 mark: Explains that attribution acknowledges the creator or source.

1 mark: Explains why a credit line does not expand reading-only permission or grant unlimited use.

1 mark: Gives a suitable next action: seek the needed permission before copying/publishing, or replace the figure with permitted material.

Q8. 4 marks

Answer: The use can follow the card because it permits sharing and adapting for any purpose. A suitable record is: "Shade icon by Map Art Team; source A-01; resized and recolored by our team." The record identifies the material and creator, retains the source identifier, and marks the actual changes.

1 mark: States that the adapted use is permitted if the given credit and change conditions are followed.

1 mark: Identifies the shade icon and Map Art Team as its creator in the record.

1 mark: Includes source identifier A-01 to make the record traceable.

1 mark: States both actual changes: resizing and recoloring.

Q9. 4 marks

Answer: For B, obtain separate copying permission or replace the figure before public reuse. For C, use the code for this project and record its author. For D, seek permission or replace it with clearly permitted material. Online availability indicates access, not the rights needed to copy, modify or redistribute.

1 mark: For B, requests the missing copying permission before reuse or replaces the figure.

1 mark: For C, permits this project's use and records the code's author.

1 mark: For D, seeks permission or replaces the photograph rather than assuming reuse is allowed.

1 mark: Explains that online availability does not itself grant copying, modification or redistribution rights.

Q10. 4 marks

Answer: Cost = 2 + 2 = 4, within budget. Eligible = 40 + 20 + 15 + 10 = 85. The 10 skills-only and 5 accessible-UI-only residents remain blocked under this selection. Eligibility does not guarantee actual participation or a representative sample.

1 mark: Calculates total cost 4 and shows it fits the budget.

1 mark: Calculates 85 eligible residents using 40 + 20 + 15 + 10, counting the double-barrier group once.

1 mark: Identifies both unresolved groups: 10 skills-only and 5 accessible-UI-only residents.

1 mark: Explains that eligibility does not guarantee participation or representativeness, with a clear distinction from the model count.

Q11. 4 marks

Answer: Sample = $0.80 \times 20\% + 0.20 \times 80\% = 32\%$. Population-weighted reference = $0.60 \times 20\% + 0.40 \times 80\% = 44\%$. The reference is 12 percentage points higher. Within-group nonresponse or selection bias, or biased wording/measurement, can remain even if the zone shares match.

1 mark: Calculates the sample estimate as 32% using the 80/20 sample weights.

1 mark: Calculates the population-weighted reference as 44% using the 60/40 population weights.

1 mark: States that the reference is 12 percentage points higher than the sample estimate.

1 mark: Explains a specific remaining within-group or measurement bias that matching zone proportions does not eliminate.

Q12. 4 marks

Answer: Collect zone and shadeRating to group and summarize need, omitting unnecessary names, emails and exact GPS. Publish zone aggregates to reduce individual detail exposure. One suitable measure is password plus verified phone approval, combining knowledge and possession to reduce unauthorized account-access risk; another is correctly applied encryption for confidentiality. Aggregates can still be linked with other data, or MFA/encryption can leave phishing, misuse or biased input unresolved.

1 mark: Chooses zone and shadeRating and connects them to grouping and measuring need while limiting unnecessary personal fields.

1 mark: Chooses an aggregate release and explains how it reduces exposure of individual details.

1 mark: Specifies a suitable authentication or encryption measure and correctly explains its protective role.

1 mark: Explains a concrete remaining risk, such as linkable aggregate data, authorized misuse, deceptive requests or biased input; does not claim zero risk.